

Prime Factorization

Lesson 1-1

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

7 has only two factors: 1×7 . So 7 is prime.

Prime number

A number bigger than 1 that you can only divide by 1 and itself.

$12 = 1 \times 12, 2 \times 6, 3 \times 4$ — six factors, so 12 is composite

Composite number

A number bigger than 1 that you can divide by more than just 1 and itself.

$36 = 2 \times 2 \times 3 \times 3 = 2^2 \times 3^2$

Prime factorization

Writing a number as prime numbers multiplied together.

$24 \rightarrow 4 \times 6 \rightarrow (2 \times 2) \times (2 \times 3) \rightarrow 2 \times 2 \times 2 \times 3$

Factor tree

A picture that splits a number into its prime numbers, step by step.

2^3 means $2 \times 2 \times 2 = 8$

Exponent

A small number that tells how many times to multiply a number by itself.

Key Ideas & Notes

- The space station received 60 supply crates.
- Mission Control needs to break this quantity into its prime components so the sorting robots can distribute them into equally sized pods.
- Help the crew find the prime factorization of 60!
- Sort these numbers — which are prime and which are composite?

Think About It

- What number are we breaking down into factors?
- What's the difference between a factor and a prime factor?
- How many different ways can we start breaking 60 apart?

My Notes

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Mission Control needs to break the 60 supply crates into prime parts. Is 60 a prime number or a composite number, and how can you tell before you even start the factor tree?

Level 1 support

Start here: 60 is composite because it has more than two factors.

60 is a ____ number because it has ____.

60 es un número ____ porque tiene ____.

I can tell before factoring because ____.

Puedo saberlo antes de factorizar porque ____.

Word bank: **prime number** **composite number** **factor** **divisible** **factor tree**

Listen for: Students say 60 is composite because it has more than two factors (it is divisible by numbers other than 1 and 60, such as 2, 3, 5, 6), so it can be broken down.

Level 2

Could Mission Control ever be asked to factor a prime number of crates, like 59?

Explain what its factor tree would look like and why.

- A prime number like 59 ____ because its only factors are ____.
- Its factor tree would ____ because ____.

EXPLORE

Look at your factor tree for 60. How did you decide which numbers to break apart first, and how do you know when to stop?

Level 1 support

Start here: You keep splitting until every branch is a prime number.

I broke ____ into ____ because ____.

Separé ____ en ____ porque ____.

I knew I was finished when every factor was ____.

Supe que terminé cuando cada factor era ____.

A prime number is ____, so I cannot break it apart.

Un número primo es ____, así que no puedo separarlo.

Word bank: prime number composite number factor factor tree

prime factorization

Listen for: Students explain they keep splitting composite branches and stop only when every leaf is prime, ending with $60 = 2 \times 2 \times 3 \times 5$.

Level 2

Two classmates started 60 as 6×10 and 4×15 . Will they get the same prime factorization? Justify why or why not.

- They ____ get the same prime factorization because ____.
- No matter how I start the tree, I always end with ____ because ____.

CONNECT

The party planner has 72 balloons. How could the prime factorization of 72 help her find every way to arrange the balloons in equal rows?

Level 1 support

Start here: Combining the prime factors of 72 builds all of its factors.

The prime factorization of 72 is _____, written with exponents as _____.

La factorización prima de 72 es _____, escrita con exponentes como _____.

She can find all the factors by _____ the prime factors.

Puede hallar todos los factores al _____ los factores primos.

Word bank: prime factorization factor exponent composite number product

Listen for: Students give $72 = 2 \times 2 \times 2 \times 3 \times 3 = 2^3 \times 3^2$ and reason that multiplying different combinations of those primes produces all factors (1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72) and thus all equal-row arrangements.

Level 2

Why does writing the prime factorization with exponents ($2^3 \times 3^2$) make it easier to count how many factors 72 has than writing it the long way?

- Exponents make it easier because _____.
- I can use the exponents to _____ instead of listing every factor by _____.

REFLECT

How is a prime number different from a composite number, and why can every composite number be written as a product of primes?

Level 1 support

Start here: Primes are the building blocks that every composite number is made from.

A prime number has _____, but a composite number has _____.

Un número primo tiene _____, pero un número compuesto tiene _____.

Every composite number can be built from primes because _____.

Todo número compuesto se puede formar con primos porque _____.

Word bank: **prime number** **composite number** **factor** **prime factorization**
product

Listen for: *Students contrast exactly-two-factors (prime) with more-than-two-factors (composite) and explain that primes are the building blocks, so any composite splits into a unique product of primes.*

Level 2

Is 1 prime, composite, or neither? Defend your answer using the definition of a prime number.

- 1 is _____ because it has _____ factor(s).
- It does not fit the definition of prime because _____.

Guided Examples

Example 1

Which of the following is a prime number?

Solution: 17 has exactly two factors: 1 and 17. $15 = 3 \times 5$, $21 = 3 \times 7$, and $9 = 3 \times 3$, so they are all composite.

Answer: A. 17

Example 2

What is the prime factorization of 30?

Solution: $30 = 2 \times 15 = 2 \times 3 \times 5$. All three factors (2, 3, 5) are prime, so $2 \times 3 \times 5$ is the prime factorization.

Answer: A. $2 \times 3 \times 5$

Example 3

What is the prime factorization of 18?

Solution: $18 = 2 \times 9 = 2 \times 3 \times 3$. Both 2 and 3 are prime, so $2 \times 3 \times 3$ is the prime factorization.

Answer: A. $2 \times 3 \times 3$

Write About the Math

The Writing Revolution

I can explain how I broke a number down using the words prime number, composite number, factor, and exponent.

1. Kernel Sentence subject + verb

Model: Prime factorization is writing a number as prime numbers multiplied together.
Factorización prima es escribir un número como números primos multiplicados.

Write a kernel sentence about prime factorization. Use a subject and a verb.

Escribe una oración base sobre factorización prima. Usa un sujeto y un verbo.

2. Sentence Expansion because • but • so

Kernel: Prime factorization matters in math
Factorización prima importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Prime factorization matters in math because ____.
Factorización prima importa en matemáticas porque ____.

but
pero

Prime factorization matters in math, but ____.
Factorización prima importa en matemáticas, pero ____.

so
entonces

Prime factorization matters in math, so ____.
Factorización prima importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about prime factorization.
Di un hecho verdadero sobre prime factorization.

Prime factorization ____.

Question *Pregunta*

Ask a question about prime factorization.
Haz una pregunta sobre prime factorization.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about prime factorization.
Muestra entusiasmo sobre prime factorization.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with prime factorization.
Dile a un compañero qué hacer con prime factorization.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

60 is a ____ **number because it has** ____.

60 es un número ____ *porque tiene* ____.

I can tell before factoring because ____.

Puedo saberlo antes de factorizar porque ____.

I broke ____ **into** ____ **because** ____.

Separé ____ *en* ____ *porque* ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Which of these numbers is composite?

- A. 27
- B. 23
- C. 29
- D. 31

Show your work:

2. Two students found different factor trees for 60. Student A started with 2×30 . Student B started with 6×10 . Which statement is true?

- A. Both get the same prime factorization: $2 \times 2 \times 3 \times 5$
- B. Only Student A gets the correct prime factorization
- C. Only Student B gets the correct prime factorization
- D. They will get different prime factorizations

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Choose any two-digit composite number. Show TWO different factor trees that both lead to the same prime factorization. Explain why every composite number has only one prime factorization.

Sentence starter: I chose the number _____. My first factor tree starts with _____ $\times$ _____, and my second starts with _____ $\times$ _____. Both give the same prime factorization: _____. This happens because _____.

Show your work:

Reflect — Exit Ticket

What is the prime factorization of 40?

- A. $2 \times 2 \times 2 \times 5$
- B. 4×10
- C. 5×8
- D. 2×20

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. $27 - 27 = 3 \times 9 = 3 \times 3 \times 3$, so it has more than two factors. 23, 29, and 31 are all prime.
2. **Try It 2:** A. Both get the same prime factorization: $2 \times 2 \times 3 \times 5$ — *The Fundamental Theorem of Arithmetic says every composite number has exactly one prime factorization. No matter how you start the factor tree, you always end with $2 \times 2 \times 3 \times 5$.*
3. **Exit Ticket:** A. $2 \times 2 \times 2 \times 5 = 40 = 2 \times 20 = 2 \times 2 \times 10 = 2 \times 2 \times 2 \times 5$. All factors (2, 2, 2, 5) are prime.

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Prime factorization is writing a number as prime numbers multiplied together.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).